Tools and APIs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected tool-contract record.

Explain the interface, trace an invalid call, audit a guarantee and design a revealing failure case.

Keep one habit: Keep exact fields, object identity, operation scope and failure status together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Conversation State and Memory. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Retrieval Augmented Generation. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Source Quality. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Conversation State and Memory

Retrieval Augmented Generation

Source Quality

W1 · Explain the tool contract

> Separate proposed call, permitted execution and checked result.

Your evidence record

- Room practice contract: exactly room_id as nonempty text, read-only, no extra fields. B capacity 12; Z measured capacity 0; other IDs absent. found=true needs matching ID and integer capacity>=0; found=false has no capacity measurement. Only transient timeout allows a second total attempt; stop after two. Separate book request uses book_id text, registry C7/M2, no lending/deletion.

  Diagram object notation is shorthand, not strict JSON.

- Reference for this exercise: open this week's student lesson, "Tools and APIs", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the extra-field request

> Reject outside-contract actions before invocation.

Your evidence record

- Room practice contract: exactly room_id as nonempty text, read-only, no extra fields. B capacity 12; Z measured capacity 0; other IDs absent. found=true needs matching ID and integer capacity>=0; found=false has no capacity measurement. Only transient timeout allows a second total attempt; stop after two. Separate book request uses book_id text, registry C7/M2, no lending/deletion.

  Diagram object notation is shorthand, not strict JSON.

- Proposed call lookup_room has room_id=B and extra **argument** book=true. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Audit a tool-success guarantee

> Success for one read-only operation is not broader authority.

Your evidence record

- Room practice contract: exactly room_id as nonempty text, read-only, no extra fields. B capacity 12; Z measured capacity 0; other IDs absent. found=true needs matching ID and integer capacity>=0; found=false has no capacity measurement. Only transient timeout allows a second total attempt; stop after two. Separate book request uses book_id text, registry C7/M2, no lending/deletion.

  Diagram object notation is shorthand, not strict JSON.

- A report omits this qualification: A successful tool result supports only its declared data and operation; it does not authorise subsequent side effects.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Design a revealing tool case

> Supply exact request, response or timeout sequence and predicted outcome.

Your evidence record

- Room practice contract: exactly room_id as nonempty text, read-only, no extra fields. B capacity 12; Z measured capacity 0; other IDs absent. found=true needs matching ID and integer capacity>=0; found=false has no capacity measurement. Only transient timeout allows a second total attempt; stop after two. Separate book request uses book_id text, registry C7/M2, no lending/deletion.

  Diagram object notation is shorthand, not strict JSON.

- Reference for this exercise: open this week’s student lesson, “Tools and APIs”, at “See how it works” and “Follow a worked example”. Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Validate the fresh K room calls independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Apply the same room input contract: exactly room_id as a nonempty string; extra fields are rejected before invocation. Treat the K values in the task's shorthand as text. K is absent from the registry. No real API, deletion capability or capacity measurement for K is supplied.

Tool accepts room_id string. Calls are {room_id:7}, {room_id:K} and {room_id:K, delete:true}. Validate; K is absent from the registry.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Allow only retrieval and calculator tools with validated **argument**s and visible results.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Trace the worked case for Tools and **API contract**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Tools and **API contract**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A tool can calculate or search supplied notes. The request is 'What is 17×6, and where do roots take up water?' Assign each subtask and describe **output** checks.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

An assistant has calculator and source-search tools. Request: 12 times 9 and habitat definition from a class pack.

Choose and verify tools.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A tool can look up book availability and also place an order. The user asks only whether a book is available. Which action is authorized?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 21 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A tool can calculate or search supplied notes. The request is 'What is 17×6, and where do roots take up water?' Assign each subtask and describe **output** checks.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Tool accepts room_id string. Calls are {room_id:7}, {room_id:K} and {room_id:K, delete:true}. Validate; K is absent from the registry.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

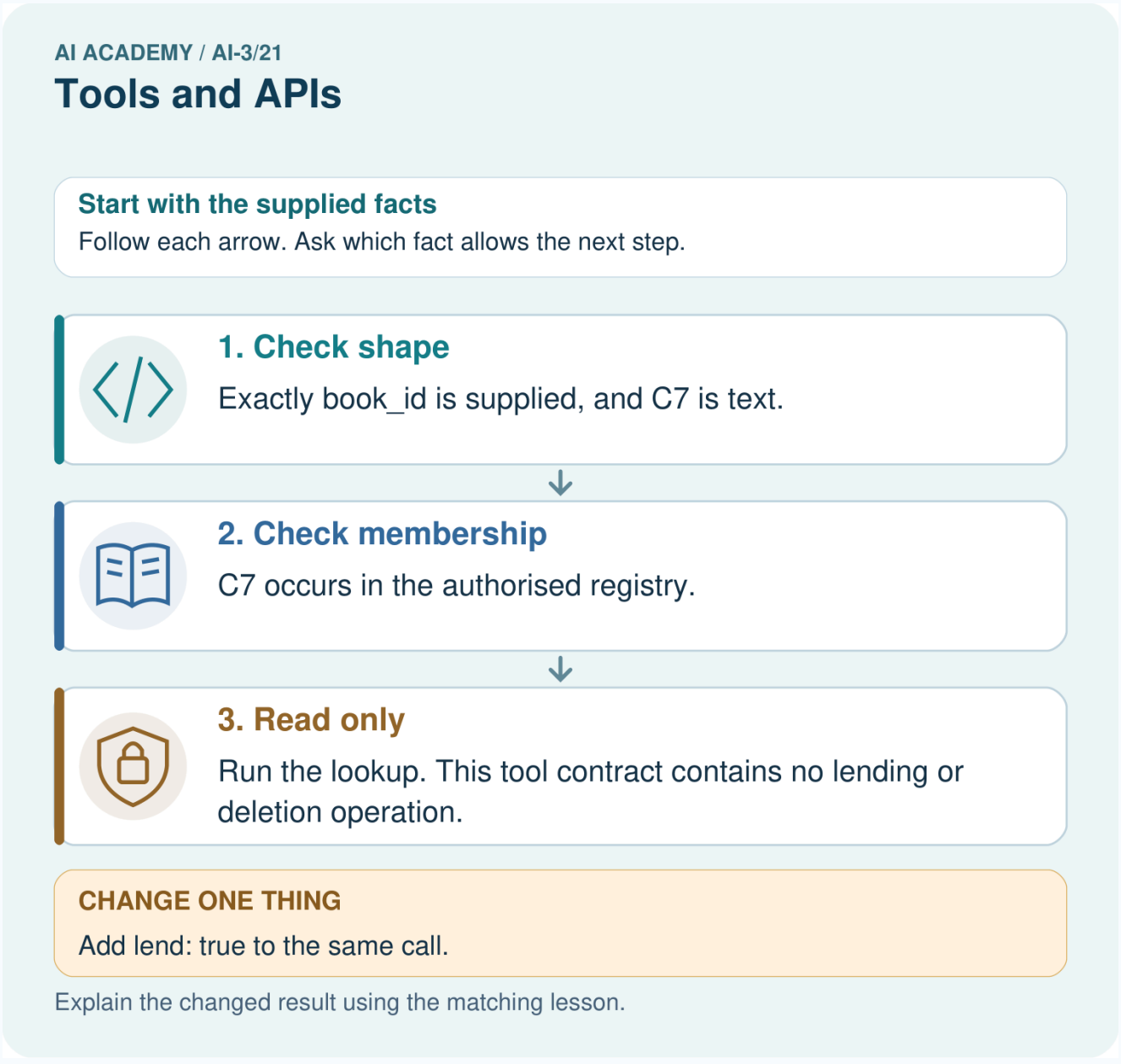
> Keep the explanation and the support you used.

Use this when helpful

Check a library lookup call before it reaches a tool.

lookup_book accepts exactly one field: book_id, a string. Registry IDs are C7 and M2. The proposed call is {book_id: C7}.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

Does a valid C7 ID permit the extra lending action? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Verified Study Buddy

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- Check a library lookup call before it reaches a tool.
- Which labelled step matches this detail? Run the lookup. This tool contract contains no lending or deletion operation.
- 1. Check shape

- 2. Read only
- 3. Check membership
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: Read only Run the lookup. This tool contract contains no lending or deletion operation. Read the picture in order. Run the lookup.

  This tool contract contains no lending or deletion operation. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |

▼ **Check the first explanation**

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: Add lend: true to the same call. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.

- 2. Reject the call before invoking the tool.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.
- After your attempt, compare with the supported explanation: Reject the call before invoking the tool. An extra operation is outside the exact allowed contract, even when book_id is valid.

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

Repair a missing building block

Repair and save your work

Review Conversation State and Memory (ai-3/20). Goal: Explain the **mechanism** and terminology of conversation state and retention. Short practice: What is selected_book after the correction and after exit?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Retrieval Augmented Generation (ai-3/18). Goal: Explain the **mechanism** and terminology of building a grounded answer. Short practice: Which part of the source supports a two-minute melting claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Source Quality (ai-3/15). Goal: Explain the **mechanism** and terminology of conflicting and time-sensitive sources. Short practice: Which source fields must be checked before answering day 9?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

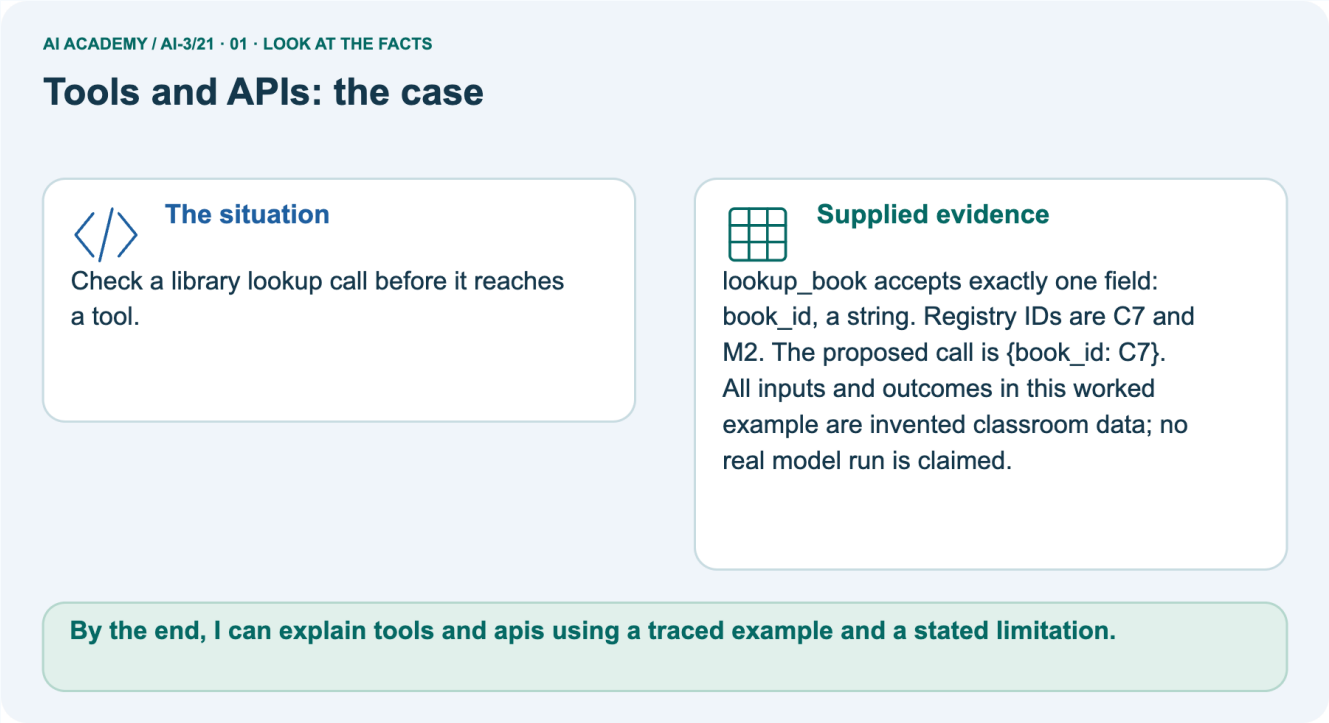
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Tools and APIs: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

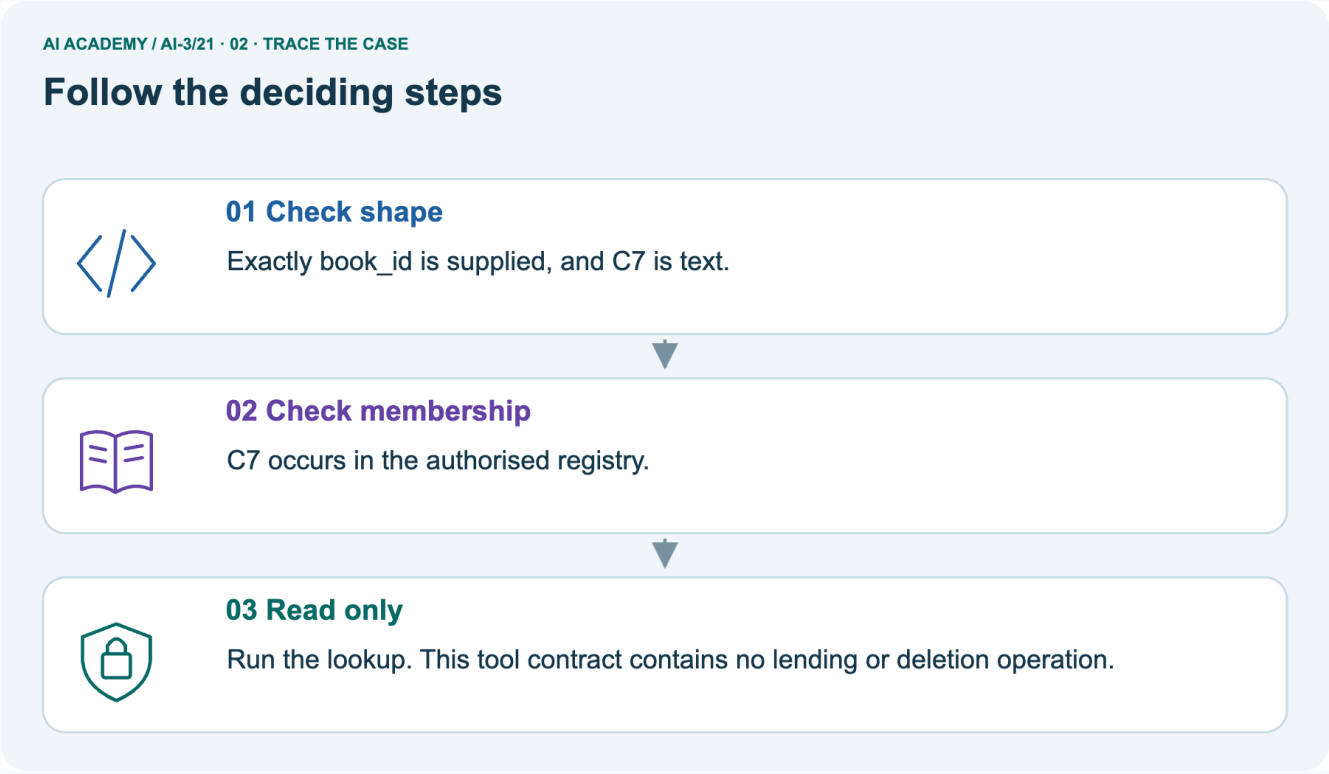
Check a library lookup call before it reaches a tool. Supplied facts: lookup_book accepts exactly one field: book_id, a string. Registry IDs are C7 and M2. The proposed call is {book_id: C7}.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



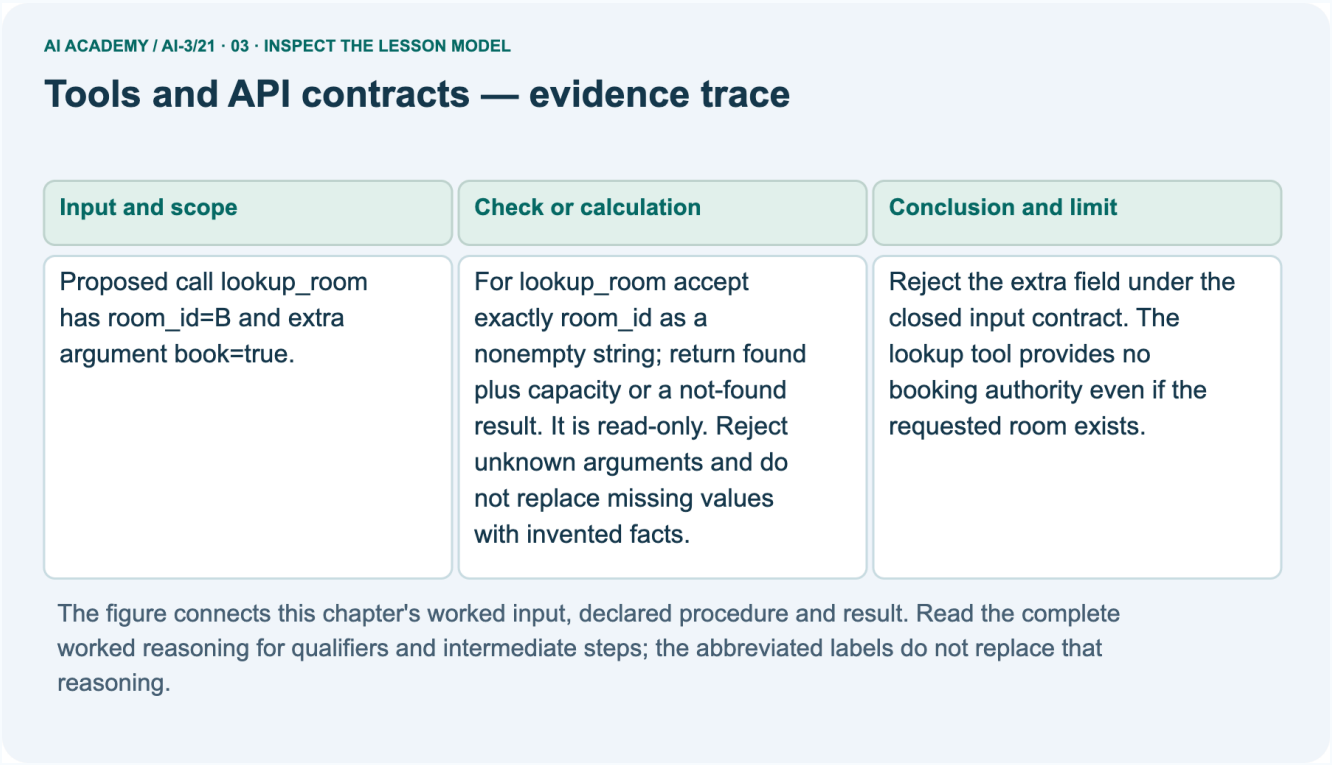
Follow the labelled picture with the matching explanation. Open the vector diagram

Check shape: Exactly book_id is supplied, and C7 is text. → Check membership: C7 occurs in the authorised registry. → Read only: Run the lookup. This tool contract contains no lending or deletion operation.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Tools and API contracts — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Tools and **API contract**s — **evidence** trace. Proposed call lookup_room has room_id=B and extra **argument** book=true. | For lookup_room accept exactly room_id as a nonempty string; return found plus capacity or a not-found result.

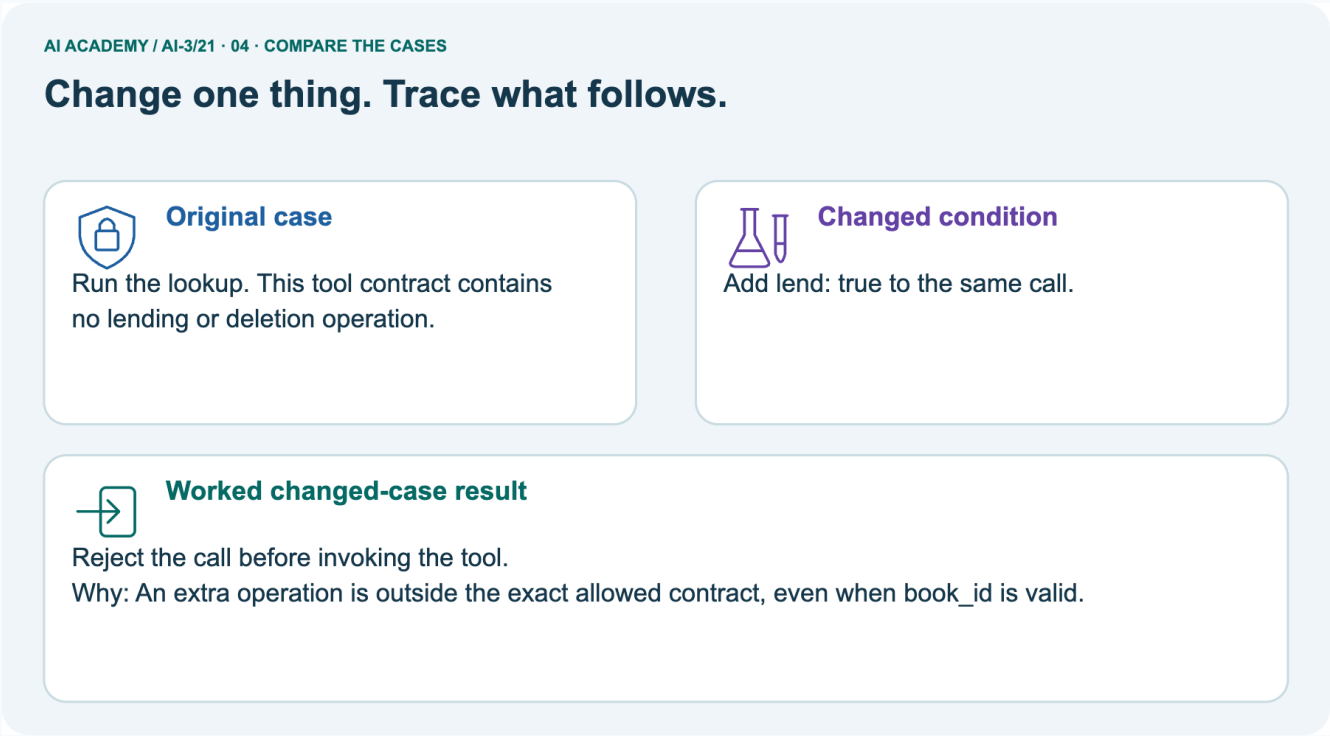
It is read-only. Reject unknown **argument**s and do not replace missing values with invented facts. | Reject the extra field under the closed **input** contract.

The lookup tool provides no booking authority even if the requested room exists.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

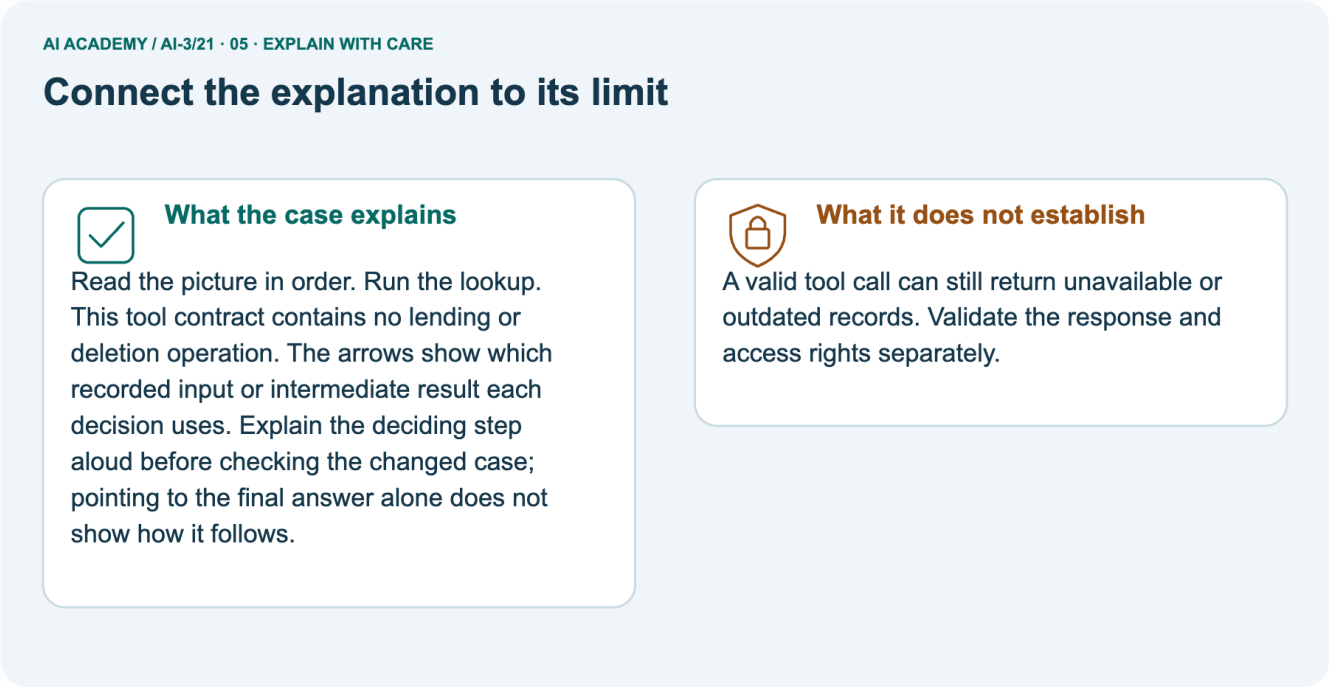
Original case: Run the lookup. This tool contract contains no lending or deletion operation. Changed condition: Add lend: true to the same call. Expected result: Reject the call before invoking the tool.

Reason: An extra operation is outside the exact allowed contract, even when book_id is valid.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Run the lookup. This tool contract contains no lending or deletion operation. The arrows show which recorded **input** or intermediate result each decision uses.

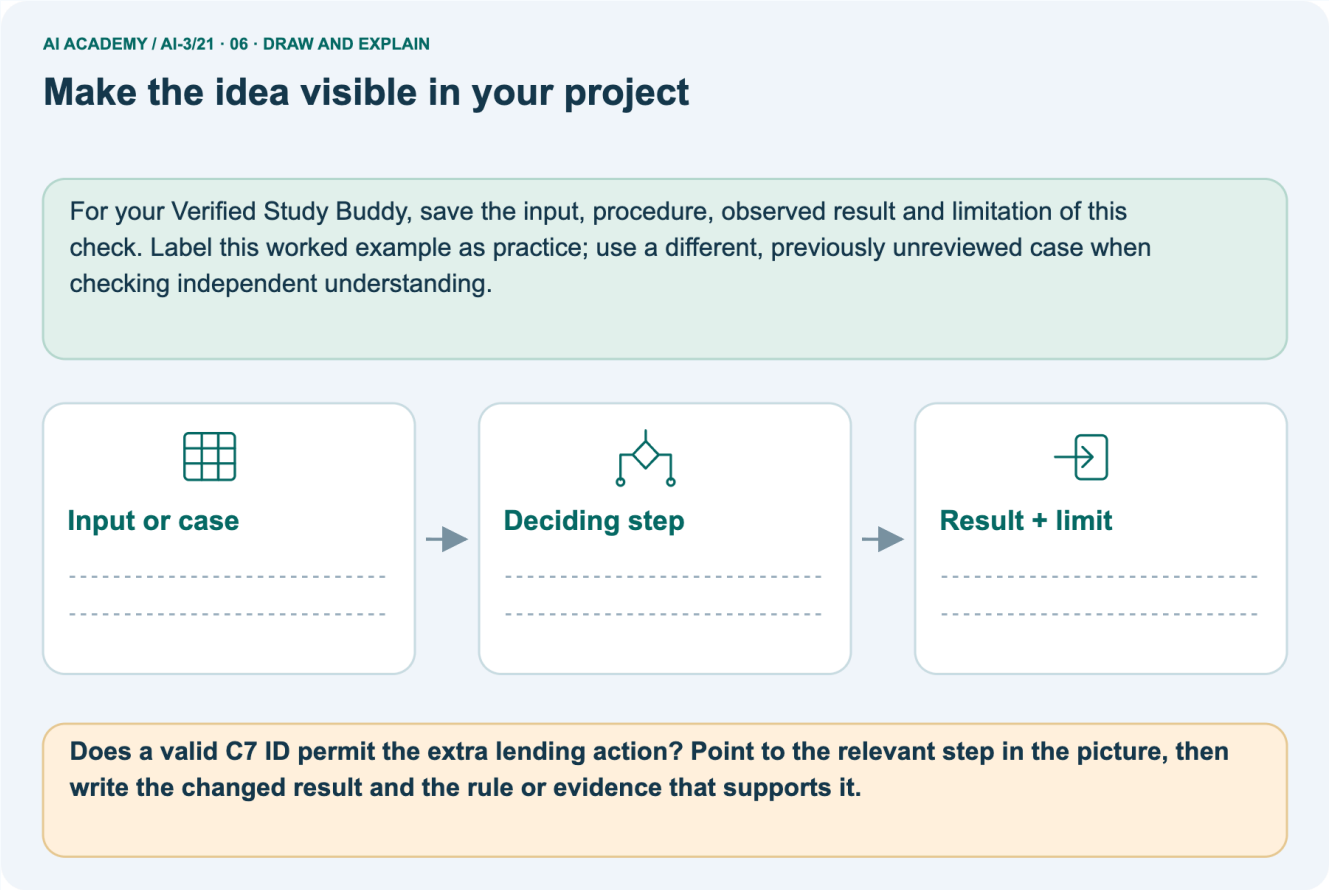
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: A valid **tool call** can still return unavailable or outdated records.

Validate the response and access rights separately.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does a valid C7 ID permit the extra lending action? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Tools and APIs: the case

02. Follow the deciding steps

03. Tools and API contracts — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.